

Exercise 4 - QuickSort Worst Case

1. Worst-Case Formula

When pivot always picks the smallest element:

$$T(n) = T(n - 1) + O(n)$$

This gives:

$$T(n) = O(n^2)$$

(Because it does $n + (n - 1) + (n - 2) + \dots + 1 = \frac{n(n-1)}{2}$ operations)

2. Worst-Case Example (16 elements)

Sorted array with first element as pivot: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16]

Step 1: pivot=1	→ left=[]	right=[2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]
Step 2: pivot=2	→ left=[]	right=[3,4,5,6,7,8,9,10,11,12,13,14,15,16]
Step 3: pivot=3	→ left=[]	right=[4,5,6,7,8,9,10,11,12,13,14,15,16]
Step 4: pivot=4	→ left=[]	right=[5,6,7,8,9,10,11,12,13,14,15,16]
Step 5: pivot=5	→ left=[]	right=[6,7,8,9,10,11,12,13,14,15,16]
Step 6: pivot=6	→ left=[]	right=[7,8,9,10,11,12,13,14,15,16]
Step 7: pivot=7	→ left=[]	right=[8,9,10,11,12,13,14,15,16]
Step 8: pivot=8	→ left=[]	right=[9,10,11,12,13,14,15,16]
Step 9: pivot=9	→ left=[]	right=[10,11,12,13,14,15,16]
Step 10: pivot=10	→ left=[]	right=[11,12,13,14,15,16]
Step 11: pivot=11	→ left=[]	right=[12,13,14,15,16]
Step 12: pivot=12	→ left=[]	right=[13,14,15,16]
Step 13: pivot=13	→ left=[]	right=[14,15,16]
Step 14: pivot=14	→ left=[]	right=[15,16]
Step 15: pivot=15	→ left=[]	right=[16]
Step 16: pivot=16	→ left=[]	right=[]

After all steps, array is fully sorted: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16]

Each step processes one element, making 16 recursion levels. Total work = $16+15+\dots+1 = \frac{16 \times 17}{2} = 136 \Rightarrow O(n^2)$ operations.

4. Results

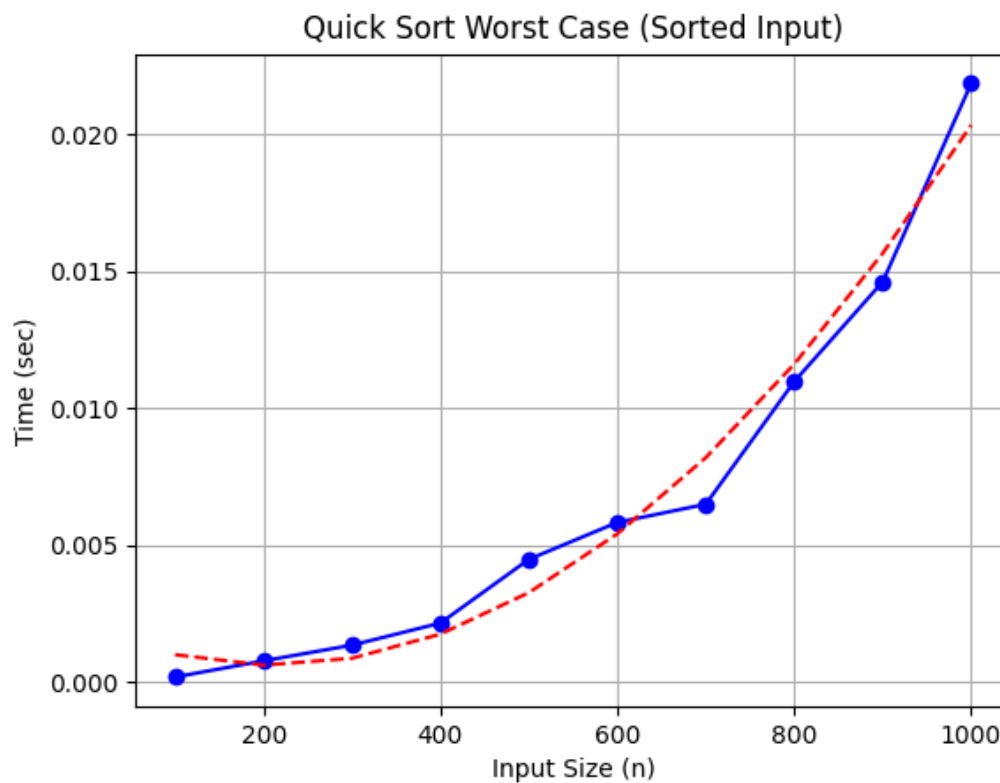


Figure 1: QuickSort on sorted arrays

Do the results match $O(n^2)$?

Yes. The curve gets steeper as n increases; When size doubles from 500 to 1000, the time increases by about 4x, confirming quadratic (n^2) growth.